

CHEEDELLA SIVA SAI JAHNAVI

Full-Stack Engineer | Generative AI & RAG

Email: cheedellajahnavi@gmail.com | Phone: +91-7981064354 | Location: Hyderabad | Open to Relocate
[LinkedIn](#) | [Portfolio](#) | [Github](#)

Full-Stack Developer with 2 years of experience delivering scalable and high-performance web applications using **React.js, Node.js, and PostgreSQL**. Experienced in building **RESTful APIs** and integrating **Generative AI** solutions such as RAG systems to automate workflows and enhance user experience. Eager to drive innovation by combining full-stack engineering with **AI-powered solutions**.

Technical Skills

- **Frontend:** React.js, Next.js, JavaScript (ES6+), TypeScript, HTML5, CSS3, Redux Toolkit (RTK), Material UI, ApexCharts
- **Backend & APIs:** Node.js, Express.js, RESTful APIs, API Integration, FastAPI
- **Databases:** PostgreSQL, Oracle Database, MongoDB, Vector Database (ChromaDB)
- **AI/ML:** Python, Generative AI, LangChain, Retrieval-Augmented Generation (RAG), NLP, Prompt & Context Engineering
- **Other Tools/ DevOps:** Git, GitHub, Linux, Firebase

Experience

Full-Stack Developer & Generative AI Engineer

TrueID, Hyderabad | October 2024 – present

- Developed scalable full-stack applications using **React.js, Node.js and PostgreSQL/Oracle DB**, improving overall system performance by 30%.
- Designed and implemented **RESTful APIs, integrating multiple backend services** to enable seamless and efficient data flow across systems.
- Optimized complex SQL queries in PostgreSQL and Oracle DB, significantly reducing response time and **improving application efficiency**.
- Built interactive data-driven dashboards with **graphs and statistics using ApexCharts**, enabling real-time insights and improving decision-making.
- Developed and integrated **Generative AI (RAG-based) solutions** to automate workflows, reducing manual effort by 50%.
- Applied **prompt engineering and context** optimization techniques, increasing AI response accuracy by 35%.
- Implemented **Redux Toolkit (RTK) and TypeScript** for scalable state management, improving code maintainability and reducing runtime errors.
- **Collaborated with Agile teams** to deliver production-ready features, while enhancing UI/UX performance and responsiveness, reducing user complaints by 40%.

React.js Intern

TrueID, Hyderabad | JULY 2024 – SEPTEMBER 2024

- Developed **reusable React components** and integrated **REST APIs** for dynamic data rendering.
- Implemented **Redux Toolkit (RTK)** for efficient state management and scalability.
- Built **responsive UI** and optimized performance, improving load times and user experience.

UI Developer Intern

BVM IT Consultancy Services, Hyderabad | APRIL 2024 – JULY 2024

- Created **responsive web pages** using HTML5, CSS3, and JavaScript with high mobile compatibility.
- Used **Flexbox, Grid, and media queries** for adaptive layouts across devices.
- Improved UI quality with **semantic HTML** and cross-browser compatibility.

Projects

IDM Dashboards | Frontend + Node.js

Technologies: React.js, Material UI (MUI), ApexCharts, Node.js, REST APIs

- Developed a dynamic analytics dashboard with reusable components and dropdown-based data switching.
- Integrated multiple APIs and implemented real-time data visualization using ApexCharts with filters like date range and operator selection.

Real-Time Messaging Platform (Glassmorphic UI) | Full stack

Technologies: React.js, Node.js, Express.js, PostgreSQL, Socket.io, JWT, Material UI

- Developed a full-stack real-time messaging platform using React.js, Node.js, and Socket.io, featuring sub-100ms message delivery and live presence tracking across concurrent user sessions.
- Implemented persistent chat history leveraging PostgreSQL and secured the application with JWT-based authentication and React Context API for seamless session and token management.

Secure Expense Tracker | Full Stack

Technologies: React.js, Node.js, Express.js, PostgreSQL

- Developed a secure expense tracking application with RESTful APIs for CRUD operations and relational data handling, along with a responsive UI using Material UI.
- Implemented Firebase Authentication with JWT-based backend verification and optimized data processing using SQL queries for filtering and aggregation.

Mind Vault | Full stack + AI

Technologies: React.js, Redux Toolkit, Python, FastAPI, Chroma DB, Gemini API

- Built a full-stack RAG-powered knowledge base using FastAPI, React.js and ChromaDB with semantic search and AI-generated answers via Gemini API.
- Implemented end-to-end ingestion pipeline - chunking, embedding, and vector storage - returning cited source snippets with every AI response.

E-Commerce web Application | Full Stack (MERN)

Technologies: React.js, Redux Toolkit(RTK), Node.js, Express, MongoDB, Socket.io

- Developed a secure e-commerce platform with authentication, product catalog, cart, and real-time inventory updates.
- Implemented backend APIs and real-time features to ensure data consistency and system reliability.

Image Labs | Full stack

Technologies: React.js, Python, PostgreSQL, Firebase

- Developed a production-grade portal to verify real vs AI-generated images with real-time backend updates.
- Scaled to support 600+ daily verifications, ensuring reliability and responsive UI performance.

RAG Q&A Chatbot | Full Stack

Technologies: React.js, Python, Flask, LangChain, FAISS, HuggingFace, Prompt Engineering

- Implemented a Retrieval-Augmented Generation chatbot to answer queries from documents, URLs, and databases.
- Achieved high accuracy across user Q&A sessions through effective prompt and context engineering.

Enterprise Frontend Systems (Merged Projects) | Frontend

Technologies: React.js, TypeScript, Redux Toolkit(RTK), Material UI

- Built and optimized multiple enterprise portals (Identity Management, Adjudication, Acquiring, Clustering) using modular React architecture.
- Integrated REST APIs to enable dynamic data rendering and seamless frontend-backend communication.
- Improved application performance and maintainability through efficient state management and component reuse.

AI Vision & Computer Vision Systems | AI/ML

Technologies: Python, YOLO, Label Studio, E2ENetworks TIR, Linux, AWS GPU

- Worked with Custom Object Training and Segmentation for Drone Imagery Using YOLO Model by Trained YOLOv11x models for complex segmentation tasks to detect objects like buildings, roads, and vehicles.
- Trained and evaluated an AI model to detect real vs AI-generated faces, including deepfakes, morphs, and face swaps. Achieved 97% testing accuracy using GPU-based training and inference on AWS across diverse datasets.

Education:

Bachelor of Technology (B.Tech) - **CGPA: 8.0/10.0**

Prakasam Engineering College

September 2020 - May 2024